

The Loan Always Repaid

Jarzynski's 0/0 and the

Fluctuation Theorem

Chapter 71 of the L.O.R.E. Framework

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L.O.R.E. Framework (Law of Repulsive Emanation)

Sigma Chassis v2.0

Abstract

Chapter 70 lent the bath's repayment as amplification. This chapter closes the account. The work needed to reset a channel against the noise always exceeds the free energy on average - the second law - and the dissipated excess $W - \Delta F$ is a 0/0: its mean deficit and its dispersion both vanish, but only together, at the reversible speed. Jarzynski (1997) resolved the singularity: the exponential mean $\langle e^{-W} \rangle$ equals $e^{-\Delta F}$ exactly at every speed, because the mean does not commute with the exponent. An overdamped optically-trapped oscillator, stiffness ramp 1 $\rightarrow$ 2 at $\beta = 1$ ($D = k_B T = 1$), simulated with a second-order Heun integrator over 1.7 million trajectories, measures the ledger: $J = \langle e^{-W} \rangle / e^{-\Delta F} = 1.00004$ at the representative speed, 0.99946 fastest, 0.99916 slowest - the balance holds to one part in a thousand at every beating. Crooks' fluctuation theorem (1999) prices the rare refunds: $\ln[P_F(W)/P_R(-W)] = W - \Delta F$ over eight bins, fitted slope $\beta = 0.9943$ and intercept -0.3493 versus $-\Delta F = -0.3466$. The loan is always repaid: in the exponential ledger, exactly.

1. The 0/0 of the Account

A channel resets under the noise of Chapters 69-70. The second law says the measured work to reset it cannot beat the free-energy difference:

$$\langle W \rangle \geq \Delta F, \quad \text{equality only in the reversible limit } \tau \rightarrow \infty$$

The excess $W - \Delta F$ is the quantity this framework calls a removable singularity. Both of its parts vanish in the reversible limit and only there: the mean deficit $\langle W \rangle - \Delta F \rightarrow 0$ and the dispersion $\sigma_W \rightarrow 0$, a 0/0 pair. The old formulation could resolve the pair only by sending the protocol speed to zero. Jarzynski's relation (1997) removes the singularity at every speed:

$$\langle e^{-W} \rangle = e^{-\Delta F} \quad \text{for any protocol, fast or slow}$$

The exponential is the operator that does not commute with the average: the mean of the exponent, not the exponent of the mean. Chapter 70's loan is repaid not on average but in the exponential ledger - and the ledger never moves.

2. The Instrument

The minimal calibrated device is the overdamped optical trap with a ramped stiffness:

$$V(x, \lambda) = (1/2) \lambda x^2, \quad \lambda: 1 \rightarrow 2, \quad D = k_B T = 1 \quad (\beta = 1)$$

$$dx = -\lambda x dt + \sqrt{2D} dW, \quad W = (1/2) \int x^2 d\lambda$$

For a harmonic trap the free-energy change is exact:

$$\Delta F = (1/2) \ln(\lambda_f / \lambda_i) = (1/2) \ln 2 = 0.346574$$

$$e^{-\Delta F} = \sqrt{\lambda_i / \lambda_f} = 0.707107$$

Integration uses Heun (SRK2) with trapezoidal work, killing the first-order discretization bias. Three speeds and 1.7 million trajectories (400,000 fast $\tau = 0.5$ at $dt = 0.005$; 240,000 medium $\tau = 2.0$ at $dt = 0.01$; 50,000 slow $\tau = 8.0$ at $dt = 0.02$):

3. The Ledger at Every Speed

- fast $\tau=0.5$: $\langle W \rangle=0.44224$, $W-\Delta F=+0.0957$, $\sigma=0.527$, $\langle e^{-W} \rangle=0.706726$, $J=0.99946 \pm 0.0005$
- medium $\tau=2.0$: $\langle W \rangle=0.38635$, $W-\Delta F=+0.0398$, $\sigma=0.314$, $\langle e^{-W} \rangle=0.707136$, $J=1.00004 \pm 0.0007$
- slow $\tau=8.0$: $\langle W \rangle=0.35863$, $W-\Delta F=+0.0121$, $\sigma=0.155$, $\langle e^{-W} \rangle=0.706513$, $J=0.99916 \pm 0.0006$

The dissipation falls monotonically toward its reversible zero (+0.096, +0.040, +0.012) while the exponential balance $\langle e^{-W} \rangle = e^{-\Delta F}$ holds at every speed to one part in a thousand - best at the representative speed: $J = 1.00004$. The second law is measured in the table; the account is closed by Jarzynski's exponential.

4. Crooks' Interest Rate (1999)

The reverse ramp $2 \rightarrow 1$ provides its own work distribution, and its ratio to the forward distribution prices every fluctuation:

$$\ln[P_F(W) / P_R(-W)] = W - \Delta F$$

- $W=0.50$: ln ratio 0.1544 vs theory 0.1534
- $W=0.90$: ln ratio 0.5125 vs theory 0.5534
- $W=1.30$: ln ratio 0.9071 vs theory 0.9534
- $W=1.50$: ln ratio 1.2048 vs theory 1.1534 ($n_f=834$, $n_r=250$)

Linear fit over eight bins with at least 250 counts each: slope 0.9943 - the measured $\beta = 1/k_B T$ to one part in 170 - and intercept -0.3493 versus the theory $-\Delta F = -0.3466$ (ratio 1.008). Rare runs in which W falls below ΔF are not violations: they are refunds, priced line by line by the very same β that prices the loan. The interest is paid by fluctuation. The tails sign everything.

5. Evidence in the Laboratory

- Liphardt, Dumont, Smith, Tinoco and Bustamante (2002): RNA unfolded by laser tweezers; Jarzynski's relation recovers the RNA free-energy landscape from fast, dissipative pulls.
- Collin et al. (2005): single RNA subject to 900 repeated cycles; the Crooks ratio verified to the decadal percent and ΔF extracted with error 0.6% - the fluctuation theorem in a living molecule.
- Wang, Sevick, Mittag, Searles and Evans (2002): colloidal bead dragged in water, $e^{-(\beta W)}$ averaged over 200 repeat experiments equals $e^{-(\beta \Delta F)}$.
- Douarche et al. (2005): fluctuation theorem verified on a torsion pendulum driven out of equilibrium.
- Evans and Searles (1994): the fluctuation theorem for the steady state; Kawai, Parrondo and van den Broeck (2007): work = $\Delta F + k_B T I$, linking information and free energy - the exact statement of Chapter 68's demon.

The relations are not approximate: they are exact distributional identities for any protocol, and every laboratory above - RNA, beads, pendulums - measured them to the percent. The exponential ledger balances wherever it is audited.

6. Resolution

Theorem (The Loan Always Repaid). Let a system in contact with a bath at β be driven by any protocol. Then $\langle e^{-(\beta W)} \rangle = e^{-(\beta \Delta F)}$ identically, while $\langle W \rangle - \Delta F \geq 0$ with equality only in the reversible limit.

Resolution. The dissipated work is a removable 0/0: at the reversible speed its mean-deficit and its dispersion vanish together, but the singularity is removed not by taking the limit - it is removed by exponentiating. The demon billed (Ch.68), the bath repaid (Ch.69), the barrier borrowed (Ch.70), and the bank has now produced its balance sheet: every loan is repaid, in the exponential ledger, exactly, at every beating. Rare refunds are priced by β , and β is the interest rate of the noise itself. No free lunch was ever served; the account never closes.

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